

■ Circuit Basics

What is a Circuit?

Grade 2 | Hardware | Science & Technology

■ Subject	Science / Hardware
■ Grade	Grade 2
■ Duration	45 minutes
■ Objective	Understand what a circuit is and how it works

■ Learning Outcomes

By the end of this lesson, students will be able to:

- ✓ Define electricity and explain its role in daily life
- ✓ Describe what a circuit is and name its four main parts
- ✓ Explain the difference between an open and a closed circuit
- ✓ Give real-life examples of circuits and follow basic safety rules

1. What is Electricity?

Electricity is the power that makes lights, fans, televisions and many other devices around us work. It is a form of energy that flows through wires and can be turned on and off.

■ Key Definition

Electricity is the flow of tiny invisible particles called electrons through a conductor (like a copper wire). When electrons move together they create an electric current that can power devices.

Things that use electricity around us:

Lights	Fans	Television
Computers	Traffic Lights	Mobile Phones

2. What is a Circuit?

A circuit is a closed path that allows electricity to flow from a power source, through connected components, and back to the source again. Think of it like a road — cars (electrons) can only keep moving if the road forms a complete loop.

■ Simple Definition

A circuit is a path that lets electricity flow to make things work.

3. Parts of a Circuit

Every simple circuit needs four key parts working together:

Component	Symbol	What it does
■ Battery	[+ –]	Provides the electrical energy (power source)
■ Wires	————	Carry electricity from one part to another

■ Bulb	(X)	Uses electricity to produce light (the load)
■ Switch	/ or —	Opens or closes the circuit to turn it on or off

4. How Does a Circuit Work?

When all four parts are connected in a complete loop, electricity can flow continuously around the path:

	Battery Produces electrical energy
	Wires Carry electricity away from the battery
	Bulb Uses the electricity — lights up!
	Back Electricity returns to the battery to complete the loop

■ Remember!

Electricity travels from the battery → through the wires → to the bulb → and back to the battery. This loop must be complete (closed) for the bulb to light up!

5. Closed Circuit vs Open Circuit

■ Closed Circuit	■ Open Circuit
The path is complete — electricity flows freely around the loop. → The bulb LIGHTS UP ■	The path is broken — electricity cannot complete the loop. → The bulb is OFF ■
Example: switch is ON	Example: switch is OFF

6. Everyday Circuits Around Us

Circuits are everywhere! They help many things work every single day:

■	Light Bulbs A simple circuit connects a bulb to electricity so it glows.
■	Ceiling Fans A motor-circuit spins the blades to create airflow.
■	Television Complex circuits process images and sound for your screen.
■	Traffic Lights Timer circuits switch red, amber and green lights automatically.
■	Mobile Phones Millions of tiny circuits fit inside to run apps and calls.
■	Computers Billions of circuits on a chip perform calculations in seconds.

7. Let's Make a Simple Circuit!

You will need: a battery, two wires and a small bulb. Follow these steps:

Step 1	Take a battery Hold a AA or 9V battery — this is your power source.
Step 2	Connect wires Attach one wire to the positive (+) end and one to the negative (–) end of the battery.
Step 3	Add a bulb Connect the other ends of the wires to the two metal contacts on the bulb holder.
Step 4	Watch it light up! If all connections are firm, the circuit is closed and the bulb will glow! ■

8. Circuit Safety Tips ■■

Electricity is very useful but can be dangerous if handled carelessly. Always follow these safety rules:

■ Don't touch wires when plugged in

Live wires can cause electric shocks. Always unplug before touching.

■ Always ask an adult for help

When working with any electrical equipment, have a grown-up present.

■ Keep circuits dry

Water conducts electricity and can cause dangerous short circuits.

■ Turn off the switch when not using

Leaving circuits on wastes energy and can cause overheating.

9. Fun Facts About Circuits! ■

■	<i>The first electric circuit was built over 200 years ago by Alessandro Volta around 1800.</i>
■	<i>Some circuits are so tiny they fit inside your mobile phone — millions on a chip!</i>
■	<i>Circuits are found in almost all electronic devices around the world.</i>
■	<i>Your brain uses tiny electric signals too — you are a living circuit!</i>
■	<i>Electricity travels through wires at nearly the speed of light.</i>

10. Review Questions

Answer these questions to check your understanding:

Q1	What is electricity?
	<i>Electricity is the flow of electrons through a conductor that produces energy to power devices.</i>
Q2	Name the four parts of a simple circuit.
	<i>Battery, Wires, Bulb (or other load), Switch.</i>
Q3	What is the difference between an open circuit and a closed circuit?
	<i>A closed circuit has a complete, unbroken path for electricity to flow (bulb lights up). An open circuit has a break in the path so electricity cannot flow (bulb is off).</i>
Q4	Give two examples of everyday devices that use circuits.
	<i>Any two from: lights, fans, TV, computer, mobile phone, traffic lights, etc.</i>
Q5	Why should we keep circuits away from water?

Water conducts electricity, which can cause short circuits and dangerous electric shocks.

■ Summary

- A circuit is a path for electricity to flow.
- Circuits need batteries, wires, bulbs and switches.
- A **closed** circuit = electricity flows = bulb lights up.
- An **open** circuit = electricity is blocked = bulb is off.
- Circuits make many things work around us every day.
- ■ Always be safe with electricity!